Frailty Indicator Analysis Platform

Progress Report: January 2026 - April 2026

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Project Title	Frailty Indicator Analysis Platform
Repository	github.com/Sreeramyapasala/Frailty-Indicator-Analysis-Platform

1. Executive Summary

This report documents the progress and technical outcomes of the machine learning internship completed at MyEdMaster LLC during the Spring 2026 semester. The primary project assigned was the development of a computer vision-based fitness assessment tool: the 30-Second Chair Stand Frailty Test, which forms a component of the My Youthspan for Athletes software platform.

The 30-Second Chair Stand Test is a lower-body strength and fall-risk assessment based on the CDC STEADI (Stopping Elderly Accidents, Deaths & Injuries) protocol. The system uses a standard webcam, the MediaPipe pose detection library, and OpenCV to guide users through a standardized stand-and-sit test, automatically count repetitions, detect arm-usage violations, and generate a CDC STEADI-compliant fall-risk score - without any physical equipment. The tool targets elderly and general adult populations and is designed for independent use at home.

Over the course of the internship, the project was developed from initial research through to a fully functional prototype - encompassing pose-based rep counting, real-time validation logic, audio guidance, a six-step position verification sequence, session history tracking, and a polished user interface aligned with company branding guidelines.

2. Project Overview

2.1 Project Title

Frailty Indicator Analysis Platform (Sit to stand 30 seconds test)

2.2 Objective

To develop a camera-based stand-and-sit assessment system that:

- Guides users through the CDC STEADI 30-Second Chair Stand Test protocol using on-screen instructions and audio prompts.
- Performs a six-step sequential body position verification before the test begins, ensuring proper setup and camera alignment.
- Automatically counts valid sit-to-stand repetitions using MediaPipe pose landmarks in real time.
- Detects arm-usage violations (arms must remain crossed on chest) and terminates the test on violation per CDC protocol.
- Generates a CDC STEADI-compliant fall-risk score categorized by age and gender across seven age groups (60-94).
- Handles error conditions gracefully including poor lighting, overexposure, partial body detection, out-of-frame events, and multiple people in frame.
- Maintains a session history log tracking results across multiple test runs within a single session.

2.3 Technology Stack

Component	Technology
Language	Python 3.7+

Pose Detection	MediaPipe (pose estimation)
Computer Vision	OpenCV (cv2)
Numerical Processing	NumPy
Audio Playback	MP3-based, cross-platform
Development Environment	Jupyter Notebook (.ipynb) + .py script
Platform	Windows / macOS / Linux
Version Control	Git / GitHub

3. Test Flow & State Machine

The application follows a fixed sequential flow. A six-step position verification sequence must be completed before the test begins, with each required position held for **2 seconds** to advance. The test itself runs for **30 seconds** from the 3-2-1-GO countdown.

#	Phase	Description
1	Basic Instructions Screen	Fullscreen "Before You Start" image displayed with 6 setup instructions read aloud via audio. User can press any key to skip each instruction or wait for audio to finish. Progress bar shown at bottom.
2	Test Instructions Slideshow	7 instruction slides shown one by one with audio narration. Each slide displays the full test instruction image on screen. For the verification version, user must hold the correct body position for 2 seconds to advance to the next slide. Spacebar skips a slide manually.
3	Starting Position Confirmation	Camera checks arms crossed on chest, seated correctly, and knees visible. Green confirmation bar shown when all three conditions are met simultaneously.
4	3-2-1-GO Countdown	Visual and audible countdown using synchronized MP3 audio with dark overlay and styled text.
5	30-Second Test	Rep counting active. Arm usage monitored every frame. Lighting and multiple-person warnings shown in top-right corner. Timer and rep count shown in styled HUD.
6	Results Display	CDC STEADI score, assessment, session history, and Retry/Quit buttons shown on a styled card overlay.

4. Development Timeline & Progress

January / February 2026 - Foundation, Research & Audio System

- Conducted comprehensive research on the CDC STEADI 30-Second Chair Stand Test protocol, including age- and gender-stratified normative score ranges and fall-risk interpretation criteria.
- Identified a key system limitation: a single test metric is insufficient for reliable fall-risk assessment without patient age and gender context.
- Implemented cross-platform MP3-based audio system replacing Windows-only dependencies; integrated countdown audio (3.mp3, 2.mp3, 1.mp3, go.mp3) and instruction audio.
- Established initial MediaPipe pose detection pipeline; reviewed existing implementation and identified architectural improvements.
- Defined team responsibility boundaries: teammate leading the Chair Sit-and-Reach flexibility test; self leading the Chair Stand frailty/fall-risk test.

March 2026 - Core Test Logic, Position Verification & UI

- Built a fullscreen Basic Instructions screen displaying the "Before You Start" poster image with aspect-ratio preservation to prevent stretching. Six instructions are read aloud one by one using non-blocking pygame MP3 playback. User can press any key at any time to skip the current audio and advance. A progress bar tracks instruction progress.
- Built a 7-slide Test Instructions slideshow with audio narration for each step. Two versions implemented: sit_to_stand_corrected.py where each slide requires the user to hold the correct body position for 2 seconds (detected via live MediaPipe pose estimation) before advancing, and sit_to_stand_no_verify.py where slides auto-advance after audio finishes with no camera verification required.
- Implemented pose detection frame skipping during instruction slides - MediaPipe runs every 2 frames instead of every frame to maintain smooth camera feed performance during the slideshow.
- Implemented camera resolution switching - resolution set to 1280x720 during instruction slides for high-quality image display, dropping to 640x480 once the test begins for real-time performance.

April 2026 - Edge Case Handling, Polish & Final Validation

- Implemented fit_image_on_canvas() utility function to display instruction images preserving their original aspect ratio with neutral padding, replacing the previous cv2.resize() which caused visible stretching and distortion.
- Fixed status message chips overflowing from the left instruction panel into the live camera side by adding a max_w boundary parameter that auto-shrinks font size to fit within the panel width.
- Completed and pushed two final versions of the software to GitHub - sit_to_stand_corrected.py (full verification) and sit_to_stand_no_verify.py (display only) - to support different clinical deployment contexts.

5. Key Technical Implementations

5.1 Rep Counting Logic

Sit-to-stand repetitions are counted by tracking the vertical Y-coordinate of hip landmarks relative to knee landmarks using MediaPipe pose estimation. A valid repetition is registered when the system detects a full standing position (hips above knees by a calibrated threshold) followed by a return to the seated position. A two-state machine (STANDING / SITTING) prevents double-counting by requiring a full position transition before incrementing the counter.

5.2 Arm Usage Violation Detection

Per CDC STEADI protocol, the participant must keep arms crossed on the chest throughout the test. The violation check monitors the Euclidean distance between each wrist landmark and the corresponding shoulder landmark every frame. If either wrist rises above the shoulder height threshold, the test immediately terminates and is marked as **Test Failed - Arm Usage Detected**, preventing invalid results from assisted stand attempts.

5.3 Six-Step Position Verification Sequence

Before the test begins, the system guides the user through six position checks displayed one by one on the live camera feed. Each step must be held correctly for 2 seconds to advance. A progress bar at the bottom of the screen shows the current step and overall progress. This sequence ensures correct camera angle, body orientation, and setup before the test is allowed to start, significantly reducing invalid attempts due to setup errors.

5.4 CDC STEADI-Compliant Scoring

The scoring system implements the full CDC STEADI normative score tables by age group (60-64, 65-69, 70-74, 75-79, 80-84, 85-89, 90-94) and gender. A below-average score for the user's age and gender group is flagged as **Increased Fall Risk**. Risk assessment is calculated only when the test completes the full 30 seconds; failed or interrupted tests display *Risk Level: N/A (Test not completed)*, preventing clinically misleading results.

5.5 Edge Case Detection

Multiple runtime conditions are detected and handled independently:

- **Dark room detection:** Grayscale mean pixel brightness checked every frame; below threshold pauses pose detection and shows a lighting warning. Auto-recovers when lighting improves.
- **Bright/overexposed room:** High pixel saturation triggers a warning; pose detection pauses until exposure normalizes.
- **Multiple people in frame:** Non-interrupting on-screen warning; test continues.
- **Out-of-frame mid-test:** Test immediately fails with a dedicated error message if user leaves frame during the 30-second window.

5.6 Session History Tracking

A session history list accumulates test results within a single application run. Each completed or failed result (rep count, risk level, pass/fail status) is appended to the list and displayed on the result screen after every run, allowing users and healthcare providers to review trends across multiple consecutive attempts without restarting the application.

5.7 Instruction Screen System

The application presents two sequential instruction screens before the test begins. The first screen displays the "Before You Start" setup guide as a Fullscreen image with six setup instructions read aloud via audio. Audio playback is non-blocking - the OpenCV window remains responsive and the user can press any key to skip the current instruction at any time. A progress bar at the bottom tracks which instruction is currently playing.

The second screen is a 7-slide instruction slideshow covering the complete test protocol: starting position, foot placement, arms position, how to stand, what counts as a valid rep, arm usage rules, and the timer. Each slide displays the full test instruction poster image on the left with the relevant instruction text overlaid at the bottom. In the verification version of the software, the right half of the screen shows the live camera feed and the user must hold each required position for 2 seconds before the slide advances. In the display-only version, slides advance automatically after audio finishes or on any keypress.

6. Edge Case Testing

A 12-scenario edge case testing document was developed and completed in April 2026. Each scenario was tested with the following documented fields: scenario description, steps taken, expected behaviour, actual behaviour, and whether the issue was fixable via code. The completed document was submitted to the supervisor for review.

Scenario	Outcome / Resolution
Basic instructions screen with audio	Complete/non-blocking MP3, keypress skip
Test instructions slideshow	Complete/7 slides, audio per slide
Live camera verification during instructions	Complete/2-second hold per step
Full body out of frame	Complete/sit_to_stand_no_verify.py
Display-only instruction version	Complete/fit_image_on_canvas()
Aspect ratio preservation for images	Documented / instruction screen guidance provided
Camera resolution switching	Complete/1280x720 instructions, 640x480 test
Pose detection frame skipping	Complete/Every 2 frames during slides

7. Known Limitations

Camera placement dependency

The test requires a front-facing camera positioned at approximately seated torso height for reliable pose detection. Non-standard camera angles may affect landmark detection reliability. This requirement is communicated through the instruction screens and visual guide.

Single-camera depth estimation

MediaPipe provides normalized 3D landmark coordinates from a single 2D camera input. Positional check accuracy may vary with camera distance and viewing angle; however, the categorical output (pass/fail per validation step) remains robust across reasonable camera distances.

Lighting sensitivity

While dark room and overexposure detection are implemented, very dynamic lighting changes (e.g., sunlight moving across the room mid-test) may cause transient detection inconsistencies. Stable, even lighting is recommended for best results.

macOS-specific OpenCV behaviours

Fullscreen keyboard events do not function reliably on macOS. All timed instruction screens use loop-based timing instead of waitKey. This is documented and functions correctly on all platforms.

8. Current Project Status

Feature	Status	Notes
Two-screen instruction display	**Complete**	*Text screen + visual image guide*
Six-step position verification	**Complete**	*2-second hold per step*
Progress bar during verification	**Complete**	*Shows current step*
Starting position confirmation	**Complete**	*Arms crossed, seated, knees visible*
3-2-1-GO countdown with audio sync	**Complete**	*MP3 playback synchronized*
30-second rep counting	**Complete**	*State machine - SITTING/STANDING*
Arm usage violation detection	**Complete**	*Immediate test failure*
Out-of-frame detection mid-test	**Complete**	*Immediate test failure*
CDC STEADI scoring (age + gender)	**Complete**	*Ages 60-94, male and female*
Risk assessment (pass-only)	**Complete**	*N/A displayed on incomplete tests*
Session history tracking	**Complete**	*Accumulates across runs*
Result screen overlay	**Complete**	*Score, risk level, history*
Dark room detection	**Complete**	*Auto-recovery on lighting fix*
Bright room / overexposure detection	**Complete**	*Warning overlay*
Multiple people in frame detection	**Complete**	*Non-interrupting warning*
Partial vs full out-of-frame check	**Complete**	*Differentiated error messages*
R key retry / Q key quit	**Complete**	*Keyboard shortcuts*
Company style guide colors applied	**Complete**	*All OpenCV screens*
Edge case testing (12 scenarios)	**Complete**	*Submitted for review*
GitHub repository	**Complete**	*Final push completed*
Demo videos	**Complete**	*Video 1 and Video 2*

9. Future Work

The following enhancements are recommended for the next development phase:

1. Audio Error Messages

Error messages are currently displayed as on-screen banners. Converting these to spoken audio alerts would significantly improve accessibility for users with visual impairments, ensuring corrective feedback is received even when the screen is not clearly visible.

2. Graphical Age and Gender Input

Patient age and gender are currently entered via terminal prompt. Migrating to an on-screen graphical input screen would eliminate the terminal dependency and enable a fully self-contained user experience.

3. Results Export to PDF or CSV

Adding the ability to save rep count, risk level, date, and session history to a PDF report or CSV file would allow users to share results with healthcare providers or wellness coaches, increasing the clinical utility of the tool.

4. Repeat Test and Best-of-N Averaging

Standard clinical practice recommends administering the Chair Stand Test two to three times and recording the best or average result. A repeat-test mode with automatic averaging would bring the prototype closer to full protocol compliance.

5. Web-Based Interface Migration

Migrating instruction screens and results display to an HTML/CSS-based web interface would enable full design system compliance, custom font rendering, and potential deployment as a browser-based tool - removing the local OpenCV installation requirement.

6. Multi-Test Platform Integration

Integrating Chair Stand Test results with the Chair Sit-and-Reach flexibility test and other frailty indicators into a unified My Youthspan dashboard would provide a more complete per-user frailty profile.

10. Conclusion

The 30-Second Chair Stand Frailty Test prototype represents a complete, functional computer vision implementation of a clinically validated fall-risk assessment protocol. The system successfully validates user positioning through a six-step verification sequence, counts sit-to-stand repetitions in real time, enforces arm-usage compliance per CDC STEADI protocol, and generates age- and gender-stratified risk scores grounded in established normative data.

The project demonstrates practical application of computer vision, pose estimation, real-time state machine design, edge case detection, and user experience design within a health and wellness context - directly aligned with MyEdMaster's mission of using data science and machine learning to make evidence-based recommendations for improving performance and wellbeing.

11. References

CDC STEADI: Centers for Disease Control and Prevention. (2017). STEADI - Stopping Elderly Accidents, Deaths & Injuries. https://www.cdc.gov/steadi

Jones & Rikli, 2002: Jones, C. J., & Rikli, R. E. (2002). Measuring functional fitness of older adults. The Journal on Active Aging, 1, 24-30.

Rikli & Jones, 1999: Rikli, R. E., & Jones, C. J. (1999). Development and validation of a functional fitness test for community-residing older adults. Journal of Aging and Physical Activity, 7(2), 129-161.

MediaPipe: Google. (2020). MediaPipe: A framework for building perception pipelines. https://mediapipe.dev